BBC micro:bit - IEEE Girls Love Tech

Steve Case

BBC micro:bit Development Board

BBC micro:bit

Python Editor

micro:bit

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Script Name

microbit program

```
# Add your Python code here. E.g.
from microbit import *


while True:
    display.scroll('Hello, World!')
    display.show(Image.HEART)
    sleep(2000)
```

BBC micro:bit

Mu Python Editor

Mu 1.0.2 - untitled *

Mode | New | Load | Save | Flash | Files | REPL | Plotter | Zoom-in | Zoom-out | Theme | Check | Help | Quit

untitled *

```
from microbit import *
while True:
    if button_a.was_pressed():
        display.scroll(temperature())
```

REPL

REPL pronounced “rep-el” stands for

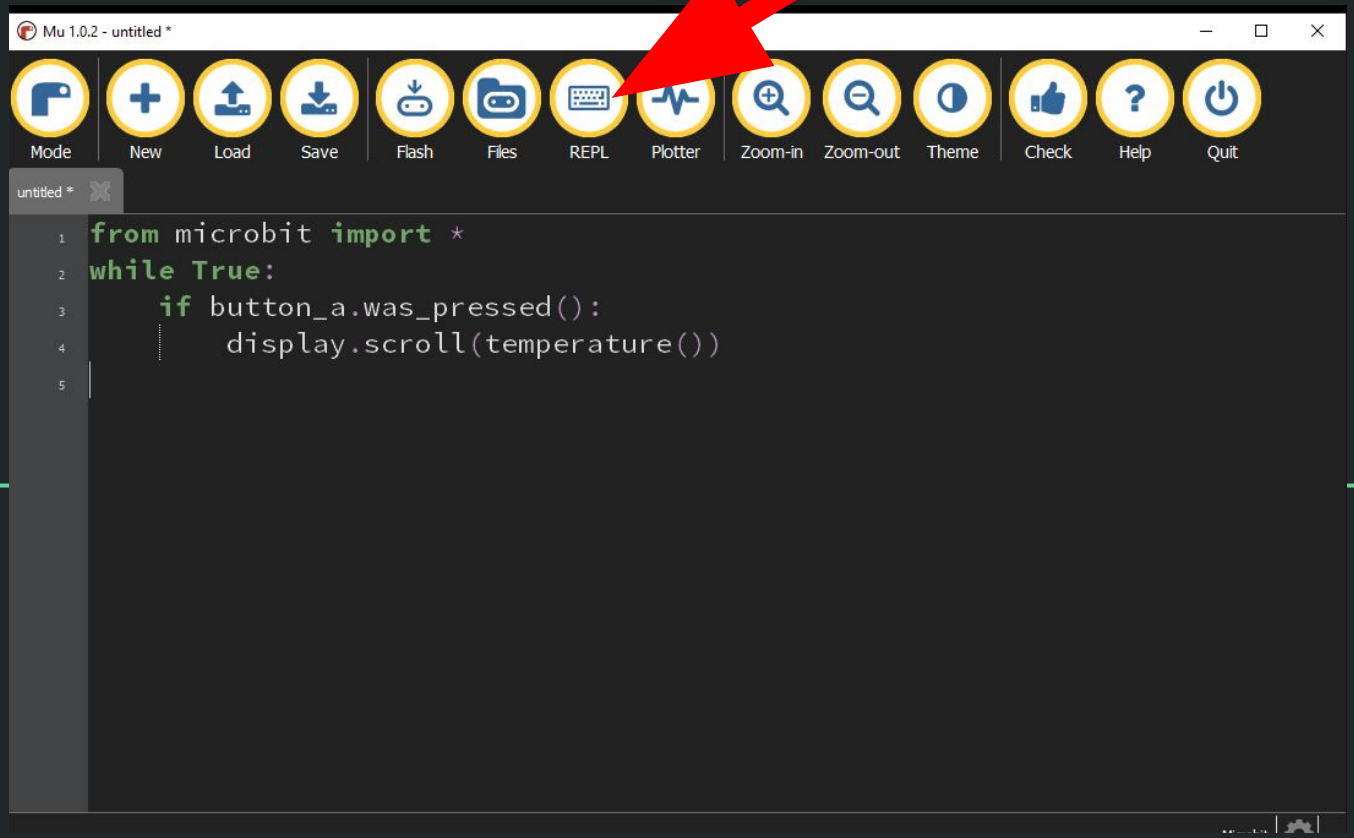
READ
EVALUATE
PRINT
LOOP

The REPL is used to communicate like a command-line to the Python interpreter.

We use it as a short-cut to test code pieces quickly before we start writing a program.

BBC micro:bit

REPL Access



BBC micro:bit

REPL is here

Mu 1.0.2 - temperature1.py

Mode | New | Load | Save | Flash | Files | REPL | Plotter | Zoom-in | Zoom-out | Them | Check | Help | Quit

temperature1.py | while_loop1.py | mu_plotter.py | for_loop1.py | dice.py | temperature2

```
# Temperature Measurement Program
#
# s.case 10/18/20
#
# The microbit has an internal thermistor on the [illegible]al microprocesor chip.
# If we use the microbit libraries, we can acc[illegible] the thermistor and the display.
#
# The microbit has internal libraries [illegible]ode stored in memory. We don't have to
# write any special code to perform [illegible]ternal fucntions. Later, we will discover
# how to find out about all the [illegible]braries present in the microbit.

from microbit impor[illegible]
while True:
    if button_a[illegible]d():
        displ[illegible].scroll(temperature())
```

BBC micro:bit REPL

```
MicroPython v1.9.2-34-gd64154c73 on 2017-09-01; micro:bit v1.0.1 with nRF51822
Type "help()" for more information.
>>>
>>>
```

Microbit

Python Basics - Variables

```
>>> i = 1
>>> print(i)
1
```

In this case, i is an integer. Think of integers as counting numbers.

```
>>> j = 2.
>>> print(j)
2.0
```

In this case, j is an floating point number. Think of it as a number with a fractional part.

```
>>> name = "Coco"
>>> print(name)
Coco
```

In this case, name is a string of collected characters.

Anatomy of bbc:microbit python programs

```
# For-Loop 1
#
# s.case 10/26/20
#
#
```

Comments

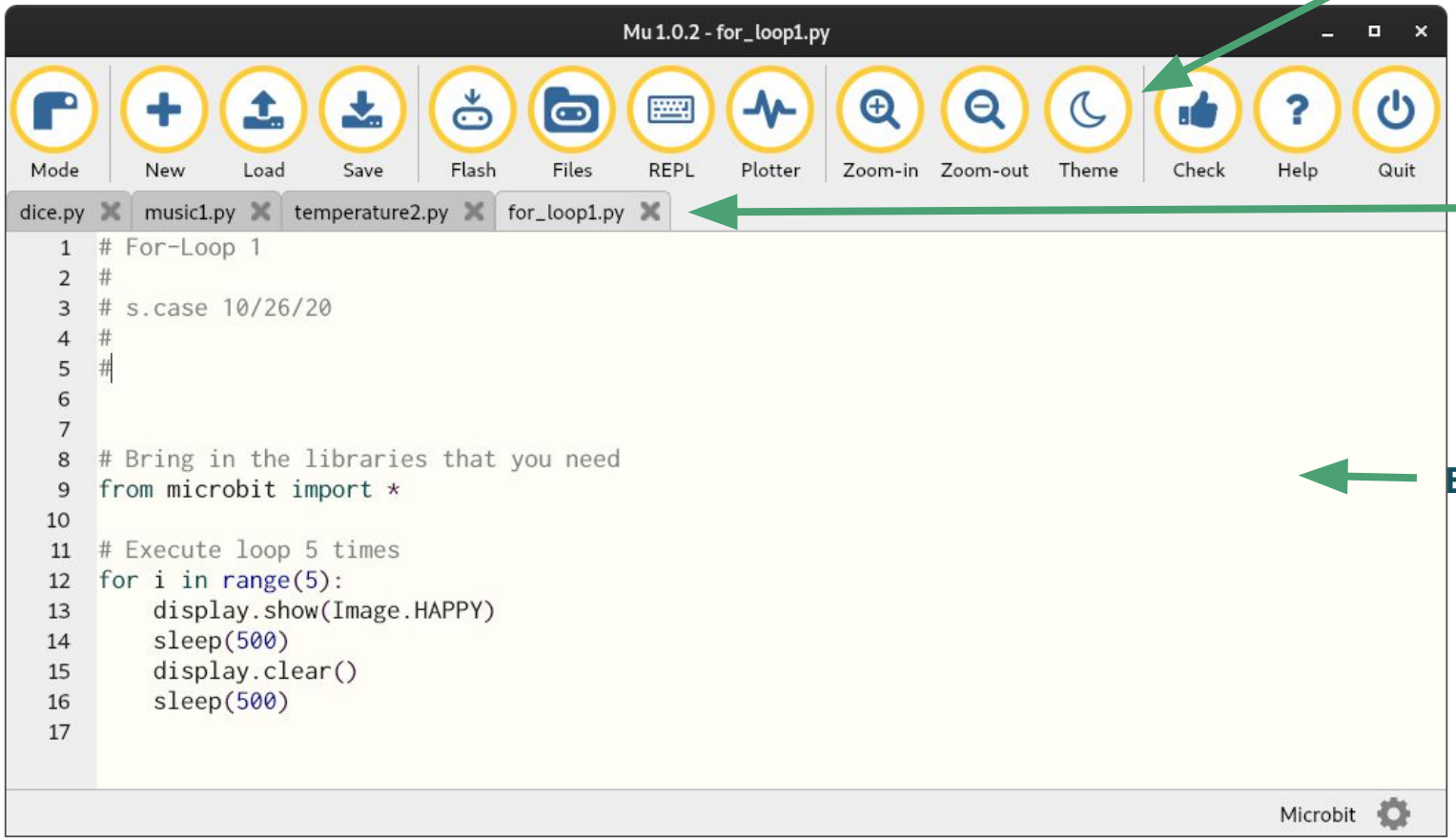
```
# Bring in the libraries that you need
from microbit import *
```

Libraries

```
# Execute loop 5 times
for i in range(5):
   display.show(Image.HAPPY)
   sleep(500)
   display.clear()
   sleep(500)
```

Running program

Mu Editor for Python



Controls

Programs on tabs

Editing area

Reference Material

Micropython Reference for the micro:bit
https://microbit-micropython.readthedocs.io/en/v1.0.1/index.html

Mu-Editor Installation:
https://codewith.mu/en/download

Seminar Slides:

Python Language:
https://www.w3schools.com/python/default.asp
https://www.python.org/ (advanced full reference for Python)
https://wiki.python.org/moin/BeginnersGuide (still pretty advanced)

Electronics Basics Vendors
Adafruit Industries: www.adafruit.com
Sparkfun electronics: www.sparkfun.com